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Hericium erinaceus ethanol extract and ergosterol exert anti-inflammatory activities by neutralizing lipopolysaccharide-induced pro-inflammatory cytokine production in human monocytes

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Abstract

Edible mushrooms are known to exert anti-inflammatory effects. In this study, the effects of ethanol extracts from edible mushrooms, such as *Hericium erinaceus*, and other edible mushrooms on inflammatory responses were investigated. Experiments were conducted using the inflammatory responses of human monocytes induced by lipopolysaccharide (LPS), a bacterial component, that provokes inflammation. Notably, we demonstrated that LPS mixed with ethanol and hot water extracts derived from edible mushrooms attenuated the production of inflammatory cytokines, such as interleukin (IL)-1β, -6, and -8, induced by LPS in human monocytic cell cultures. Moreover, we found that the ethanol extract of *H. erinaceus* contained ergosterol, which attenuated IL-8 production in LPS-stimulated cells. Subsequent component analysis of the ethanol extract of *H. erinaceus* revealed that ergosterol binds to lipid A to attenuate LPS-induced inflammation. Together, our findings suggest that ergosterol in ethanol extracts from edible mushrooms can prevent the induction of inflammation by binding to LPS.

Keywords: edible mushroom, *Hericium erinaceus*, ethanol extract, lipopolysaccharide, anti-inflammatory activity

Abbreviations:

lipopolysaccharide (LPS)

maxacalcitol (OCT)

enzyme-linked immunosorbent assay (ELISA)

Hericium erinaceus ethanol extracts (HEE)

Gas chromatography-mass spectrometry (GC-MS)

Toll-like receptor (TLR)

1. Introduction

Edible mushrooms have long been used as medicinal commodities because they contain over 2000 nutrients that are generally beneficial to health [1,2]. Mushroom-derived extracts exhibit various beneficial effects against inflammation, infections, and cancer [1,3]. Previous studies have demonstrated that mushrooms contain numerous natural compounds with anti-inflammatory activities [3-5], including polysaccharides (β-glucan, trehalose, and chitosans), vitamins, phenolic compounds, indole compounds, mycosteroids, fatty acids, carotenoids, folic acid, and bioelements (zinc, selenium, and copper) [3,6]. These studies reporting the beneficial effects of mushroom-derived components primarily focused on the efficacy of mushroom extracts in humans. However, only a few studies have assessed the interaction between mushroom compounds and inflammatory substances, such as bacterial-derived virulence factors.

An edible medicinal mushroom, *Hericium erinaceus*, is widely used for cooking owing to its health benefits and nutritional values. *Hericium erinaceus*-derived extracts contain polysaccharide, erinacine, hericerin, hericenone, resorcinol, sterol, vitamin B_{12}, and volatile compounds and ergosterol and erinacine can be extracted from *H. erinaceus* fruiting bodies [7]. In particular, *H. erinaceus* attenuates inflammatory responses in gastrointestinal diseases, Crohn's disease, and ulcerative colitis [8-10]. Thus, *H. erinaceus* has been investigated for its various beneficial health properties, and the numerous bioactivities of *H. erinaceus* may potentially have similar effects on various medical conditions.

During the inflammatory response, macrophages perform three functions: inflammation, phagocytosis, and antigen presentation. In addition, macrophages are a major source of

inflammatory mediators, including interleukin (IL)-1β, -6, and -8, nitrite oxide, and lipopolysaccharide (LPS). LPS is a bacteria-derived component that is an inducer of inflammatory mediators [11]. Attenuation of LPS-induced macrophage activation has therefore been suggested as an important target for preventing inflammatory diseases. Interestingly, previous studies have reported that *H. erinaceus*-derived ethanol and ethyl acetate extracts can suppress LPS-induced inflammatory responses by macrophages [12,13]. Moreover, these studies have suggested that *H. erinaceus* extracts attenuate LPS-induced activation of host cells by inhibiting signaling molecules, such as nuclear factor-κB and mitogen-activated protein kinase, which can induce the expression of pro-inflammatory cytokines.

In this study, we investigated the anti-inflammatory activity of edible mushroom extracts by directly interacting with LPS, using pro-inflammatory cytokine production by human THP-1 monocytic cells as an indicator. We used the following edible mushrooms, which are commonly available in Japan: *H. erinaceus, Grifola frondosa* (ram's head or Japanese "maitake"), *Pleurotus eryngii* (king oyster mushroom or Japanese "eringi"), *Lentinus edodes* (shiitake mushroom), *Lyophyllum shimeji* (Japanese "hon-shimeji"), and *Agaricus blazei*. We hypothesized that the ethanol extracts of edible mushrooms could neutralize inflammatory activity by binding to LPS and prevent LPS-induced inflammation in human THP-1 monocytic cells by neutralizing the inflammatory activity of LPS.

2. Materials and Methods

2.1. Mushroom extract preparation

Ethanol and hot water extracts from *H. erinaceus*, *G. frondosa*, *P. eryngii*, *L. edodes*, *L. shimeji*, and *A. blazei* were kindly provided by Hokuto Corporation (Nagano, Japan). Briefly, the fruiting bodies of each mushroom were lyophilized, homogenized, and extracted with ethanol at 25 °C for three days. These extracts were filtered, evaporated, and used as ethanol extracts. Fresh fruiting bodies of the same mushroom species were also homogenized and extracted with the same volume of distilled water at 100 °C for 10 min. Following cooling to 25 °C, the extracts were filtered, lyophilized, and used as hot water extracts.

2.2. Reagents

LPS from *Escherichia coli* O55:B5 and a synthetic muramyl dipeptide were obtained from Merck (Darmstadt, Germany). Synthetic *E. coli*-type lipid A (LA-15-PP), the minimal active structure of LPS, was obtained from the Peptide Institute Inc. (Osaka, Japan). FSL-1, a synthetic diacylated lipoprotein, and *Porphyromonas gingivalis* LPS were obtained from InvivoGen (San Diego, CA, USA). LPS was purified from the *Fusobacterium nucleatum* JCM 8532 strain, as described previously [14]. Peptidoglycan was purified from *Levilactobacillus brevis* NBRC 3345, as described previously [15]. Standard products of ergosterol was obtained from FUJIFILM Wako Pure Chemical Industries (Osaka, Japan), and ergosterol peroxide and 1α,25-dihydroxyvitamin D_2 (ercalcitriol) were obtained from Cayman Chemical (Ann Arbor, MI, USA). Unless otherwise indicated, all remaining reagents were from Merck.

2.3. Cell culture

The human monocytic leukemia cell line THP-1 (Japanese Collection of Research Biosources Cell Bank, Hokkaido, Japan) was grown in RPMI 1640 medium (Nacalai Tesque, Kyoto, Japan) with 10% heat-inactivated fetal bovine serum (FBS; HyClone, Logan, UT, USA) and 1% antibiotic-antimycotic agents. Undifferentiated THP-1 cells lacking CD14 (LPS receptor) expression were treated with 0.1 μM maxacalcitol (22-oxacalcitriol: OCT), a synthetic vitamin D_3 analog, for three days to induce differentiation into monocyte-like cells that strongly express CD14 on the cell membrane [16].

The viability of OCT-differentiated THP-1 cells treated with mushroom ethanol extracts was measured using the RealTime-Glo™ MT cell viability assay (Promega, Madison, WI, USA) according to the manufacturer's instructions. Briefly, cells treated with mushroom-derived extract were incubated with cell membrane-permeable NanoLuc® luciferase substrate reduced to a NanoLuc® luciferase substrate by viable cells. When the NanoLuc® substrate diffuses out into the culture medium, it is converted to a luminescent signal by NanoLuc® luciferase. Viable cells in the presence of the assay reagent constantly produce luminescent signals. The luminescence of the culture supernatant was measured using a GloMax® Explorer Multimode microplate reader (Promega).

2.4. Measurement of cytokine production

Concentrations of IL-1β, -6, and -8 in the culture supernatants were evaluated using an enzyme-linked immunosorbent assay (ELISA) kit (BD Biosciences, Franklin Lakes, NJ, USA) according to the manufacturer's instructions. To induce mature IL-1β production, cells

were treated with 5 μM nigericin 1 h before the end of culture. The cytokine concentrations were assessed using SoftMax Pro software (Molecular Devices, Menlo Park, CA, USA).

2.5. Thin-layer chromatography (TLC)

Ethanol extracts of *H. erinaceus* were subjected to TLC analysis using silica gel 60 pre-coated F254 plates (Merck) and a solvent system containing acetate, chloroform, methanol, and water (90:10:1:1, v/v/v/v). Detection was performed by spraying 10% H_2SO_4/ethanol and heating.

2.6. Gas chromatography-mass spectrometry (GC-MS)

Hericium erinaceus ethanol extracts (HEE) were acetylated with acetic anhydride/pyridine (1:1, v/v) (100 °C for 30 min). Electron impact mass spectra were obtained using a GC-MS instrument (QP2010Plus, Shimadzu, Kyoto, Japan) equipped with a capillary column (25 m CBP1, Shimadzu) under the following conditions: column heating program from 50 to 300 °C, ionizing voltage 70 eV, ionization temperature 200 °C, interface temperature 300 °C, injection port temperature 300 °C, and helium gas pressure 150 kPa.

2.7. Statistical analysis

Data are presented as the mean ± standard deviation, and experiments were performed in triplicate to ensure reproducibility. Statistical analysis was performed using Student's *t*-tests, and statistical significance was set at $p < 0.05$.

3. Results

3.1. Pre-mixing edible mushroom ethanol extracts with LPS attenuate pro-inflammatory cytokine production induced by LPS in human monocytic cells

First, we evaluated the effect of ethanol extracts from edible mushroom fruiting bodies on pro-inflammatory cytokine production by *E. coli*-derived LPS in human THP-1 monocytic cells differentiated using the synthetic vitamin D_3 analog, OCT (hereafter referred to as dTHP-1 cells). First, we used three different methodologies to examine the effects of the ethanol extract of *H. erinaceus* (hereafter as HEE) on IL-8 production by dTHP-1 cells: (1) dTHP-1 cells were treated with HEE, washed, and then stimulated with LPS (Fig. 1A); (2) dTHP-1 cells were simultaneously stimulated with LPS and HEE (Fig. 1B); (3) dTHP-1 cells were stimulated with LPS that had been pre-mixed with HEE (Fig. 1C). IL-8 production levels in dTHP-1 cells induced by *E. coli* LPS were not changed in the cells pre-treated with HEE (Fig. 1A). Simultaneous stimulation of dTHP-1 cells with *E. coli* LPS with HEE partially attenuated IL-8 production by *E. coli* LPS in dTHP-1 cells (Fig. 1B). In contrast, pre-mixing HEE with *E. coli* LPS drastically reduced LPS-induced IL-8 production in dTHP-1 cells (Fig. 1C). These results raised the possibility that the ethanol extract derived from edible mushrooms could be bound to LPS, resulting in the attenuation of IL-8 production induced by LPS in human monocytic cells. We therefore performed experiments with various edible mushrooms. Ethanol extracts from edible mushrooms did not induce IL-8 production in the dTHP-1 cells (Fig. 1D); however, simultaneous stimulation of dTHP-1 cells with *E. coli* LPS and ethanol extracts from *H. erinaceus* and *L. shimeji* partially and significantly reduced IL-8 production (Fig. 1B). Pre-mixing *E. coli* LPS with the mushroom

ethanol extracts also significantly reduced LPS-induced IL-8 production (Fig. 1D) and completely impaired the LPS-induced IL-6 and -1β production (Fig. 1C and D) in dTHP-1 cells, indicating that mushroom ethanol extracts may disturb LPS-induced host cell activation. We also determined the efficacy of various concentrations of mushroom ethanol extracts on IL-8 production and found that pre-mixing *E. coli* LPS with 10 μg/mL HEE could significantly reduce IL-8 production by *E. coli* LPS (Fig. 1G). Moreover, we revealed that the ethanol extracts did not reduce the viability of dTHP-1 cells, indicating that cytotoxicity was not responsible for the reduction in IL-8 production (Fig. 1H). In addition, we found that pre-mixing *E. coli* LPS with lyophilized whole preparations from *H. erinaceus* fruiting bodies significantly attenuated IL-8 production by LPS in dTHP-1 cells (Fig. 1I).

3.2. Pre-mixing edible mushroom hot water extracts with LPS attenuate pro-inflammatory cytokine production induced by LPS in human monocytic cells

Next, we investigated the effect of pre-mixing hot water extracts from edible mushroom fruiting bodies with LPS on pro-inflammatory cytokine induction by LPS in dTHP-1 cells. When dTHP-1 cells were simultaneously stimulated with *E. coli* LPS and hot water extracts from *G. frondosa* and *P. eryngii,* IL-8 production induced by LPS was partially and significantly reduced (Fig. 2A). In contrast, pre-mixing *E. coli* LPS with the hot water extracts strongly reduced IL-8 production (Fig. 2A) as well as LPS-induced IL-6 and -1β production (Fig. 2B and C). In particular, hot water extract of *G. frondosa* potently repressed the production of IL-8, -6, and -1β (Fig. 2A–C). These results indicate that mixing *E. coli* LPS with ethanol or hot water extracts of edible mushrooms significantly attenuated the

inflammatory response induced by LPS, suggesting that the extracts may directly inhibit the biological activity of LPS.

3.3. Binding of the HEE to LPS inhibits IL-8 production by LPS in human monocytic cells

Lipid A is a covalently bound component of LPS [17] that is responsible for the biological activity of inflammatory responses to LPS; therefore, we examined whether ethanol and hot water extracts from edible mushrooms could attenuate lipid-A-induced IL-8 production in dTHP-1 cells. As shown in Figure 3A, pre-mixing synthetic *E. coli*-type lipid A with the ethanol extracts of edible mushrooms considerably reduced IL-8 production in dTHP-1 cells. In particular, pre-mixing HEE with lipid A strongly inhibited lipid A-induced IL-8 production (Fig. 3A). In contrast to the result with LPS (Fig. 2A), pre-mixing lipid A with hot water extracts from the edible mushrooms led to only slight inhibition in lipid A-induced IL-8 production (Fig. 3B). Furthermore, HEE exhibited inhibitory effects on IL-8 production at levels similar to those of lipid A and LPS (Fig. 3C), suggesting that ethanol extracts from edible mushrooms may bind to lipid A. Together, these results suggested that edible mushroom ethanol extracts mixed with LPS can inhibit lipid A-induced IL-8 production.

After demonstrating that the HEE mixed with LPS or lipid A could attenuate the production of IL-8 in dTHP-1 cells (Fig. 1D and 3A), the reduction in IL-8 production caused by HEE might not be due to the interaction between the extract and host cells. We then examined whether these anti-inflammatory effects were achieved by the ethanol extract interacting with the host cells or LPS itself. As the activation of host cells by LPS requires the

formation of a complex between LPS and LPS-binding protein (LBP) present in the serum [18], we stimulated dTHP-1 cells with *E. coli* LPS pre-mixed with HEE in the absence of LBP under serum-free conditions. Interestingly, pre-mixing HEE with *E. coli* LPS inhibited IL-8 production induced by LPS even in the absence of LBP (Fig. 3D). LBP-bound LPS binds to CD14 expressed on host cells and generates an activation signal that is transmitted via the Toll-like receptor (TLR) 4. It is known that high concentrations of LPS can activate the CD14-independent pathway [19]. To determine whether the HEE bound to CD14, we stimulated intact THP-1 cells lacking CD14 expression with 20 μg/mL of *E. coli* LPS (Fig. 3E). Consistently, pre-mixing *E. coli* LPS with HEE significantly reduced the induction of IL-8 production by LPS (Fig. 3E), indicating that HEE may inhibit IL-8 production by binding to LPS. Furthermore, pre-mixing HEE with LPS derived from *P. gingivalis* or *F. nucleatum*, a periodontal pathogen, significantly reduced IL-8 production by LPS in dTHP-1 cells (Fig. 3F). Although CD14-mediated cellular activation can be triggered by both LPS and peptidoglycan, a component of bacterial cell wall [20], IL-8 production was not attenuated when HEE was pre-mixed with TLR2-agonistic peptidoglycan, TLR2/6-agonistic FSL-1, or nucleotide-binding oligomerization domain (NOD)2-agonistic muramyl dipeptide (Fig. 3G). These results suggest that binding of HEE to LPS inhibits the inflammatory responses induced by LPS in human monocytic cells.

3.4. Ergosterol from the HEE attenuates IL-8 production induced by LPS in THP-1 cells

Finally, we attempted to determine the specific components in the ethanol extracts of edible mushrooms that can neutralize LPS and inhibit LPS-induced inflammation. The most

abundant component in edible mushroom ethanol extracts is ergosterol, a vitamin D_2 precursor [21-23]. Preheating HEE (30 min, 95 °C) before mixing it with LPS did not abolish its inhibitory effects of IL-8 production by LPS (Fig. 4A). Since ergosterol has been reported to exhibit heat resistance [23,24], these results strongly suggest that ergosterol could be an active component of HEE that inhibits the inflammatory activity of LPS. Therefore, we analyzed the components of HEE using TLC and GC-MS. As shown in Figure 4B, the extract contained a lipid with the same mobility as the ergosterol control (indicated by the arrow). Moreover, when HEE was acetylated, GC-MS analysis indicated that the mass spectra and time of retention of the acetylated lipid detected at approximately 300 °C were identical to those of the acetylated ergosterol control (Fig. 4C). Together, these results indicate that the HEE contained ergosterol. Indeed, IL-8 production induced by LPS from *E. coli*, *P. gingivalis*, and *F. nucleatum* was significantly reduced in cells treated with pre-mixed LPS and pure ergosterol standard (Fig. 4D). In addition to ergosterol, mushrooms contain ergosterol peroxide [25]. Furthermore, ergosterol present in mushrooms is converted to vitamin D_2 upon UV irradiation [26]. Therefore, we next examined the LPS-neutralizing effects related to ergosterol, ergosterol peroxide, and the biologically active metabolite of vitamin D_2, 1α,25-dihydroxyvitamin D_2 (ercalcitriol). IL-8 production induced by *E. coli* LPS was completely diminished in the cells stimulated by the pre-mixed LPS with ergosterol peroxide or ercalcitriol as well as ergosterol (Fig. 4E), indicating that not only ergosterol but also ergosterol peroxide and ercalcitriol in the edible mushrooms probably inhibit IL-8 production by LPS in dTHP-1 cells. Thus, these findings suggest that ergosterol in HEE

binds to LPS, resulting in the attenuation of LPS-induced pro-inflammatory cytokine production.

4. Discussion

In this study, we demonstrated the anti-inflammatory effects of ethanol and hot water extracts derived from various edible mushrooms that are commonly consumed in Japan, including *H. erinaceus, G. frondosa*, *P. eryngii*, *L. edodes*, and *L. shimeji.* We used LPS to induce an inflammatory response in human monocytic cells and found that both the ethanol and hot water extracts of edible mushrooms attenuate LPS-induced IL-1β, -6, and -8 production. Moreover, pre-mixing ethanol or hot water extracts with LPS drastically attenuated pro-inflammatory cytokine production, and the ethanol extracts also inhibited lipid A-induced IL-8 production. In particular, HEE attenuated IL-8 production by lipid A, a component involved in the inflammatory response to LPS. Furthermore, we found that pre-mixing HEE with LPS could inhibit IL-8 production by LPS in the absence of LBP and in the intact THP-1 cells lacking CD14 expression. These findings suggest that HEE may have reduced IL-8 production in human monocytic cells by binding directly to LPS. TLC and GC-MS analyses further revealed that ergosterol is the major component of HEE. The findings of this study imply that ergosterol contained in mushrooms may play a role in preventing inflammation induced by gram-negative bacteria contaminated with food products.

The major sterol found in cultivated edible mushrooms, such as *A. blazei*, *Pleurotus ostreatus*, and *L. edodes*, is ergosterol [21]. As expected, pre-mixing *E. coli*-derived LPS

with the ethanol extract of *A. blazei* completely eliminated LPS-induced IL-8, IL-6, and IL-1β production in dTHP-1 cells (supplemental fig. 1). Kawai *et al.* reported that ergosterol in the ethyl acetate extract of *G. frondosa* prevents allergic responses by inhibiting mast cell degradation [27]. When exposed to ultraviolet light at wavelengths of 280−320 nm, ergosterol is converted into vitamin D_2 ,which plays a vital role in calcium metabolism and bone mineralization [21]. Previous studies have reported that sterols are mainly present in HEE rather than its water extract [22]. Consistently, we found that HEE contained ergosterol (Fig. 4B and C) and that its inhibitory effect on IL-8 production by LPS was not abolished even when preheated (30 min, 95 °C; Fig. 4A). Together with the finding that whole preparations of *H. erinaceus* fruiting bodies reduced LPS-induced IL-8 production (Fig. 1G), our results suggest that the fruiting bodies of *H. erinaceus* may still have LPS-neutralizing properties even after cooking.

In the present study, we used three different methodologies to examine the effects of ethanol extracts of edible mushrooms on pro-inflammatory cytokine production (Fig. 1A–C): (1) dTHP-1 cells were treated with HEE, washed, and then stimulated with LPS; (2) dTHP-1 cells were simultaneously stimulated with LPS and HEE; (3) dTHP-1 cells were stimulated with LPS that had been pre-mixed with HEE. Using the first method allowed HEE to contact either CD14 or LBP first, whereas by the second method, HEE was able to simultaneously interact with LPS, LBP, and CD14 expressed on the host cell. In contrast, when the third method was used, HEE encountered both LPS and LBP before CD14. During cellular responses to LPS mediated by CD14, LPS and LBP in the serum form a complex that binds to CD14, thereby inducing inflammatory responses in host cells [18]. Herein, we found that

pre-mixing ethanol extracts with LPS markedly reduced IL-8 production in OCT-treated CD14-bearing dTHP-1 cells in the presence of LBP (Fig. 1D), in the absence of LBP (Fig. 3D), and in THP-1 cells lacking CD14 expression (Fig. 3E). Together, these results suggest that the ethanol extract likely binds to LPS, but not to LBP or CD14. The TLR2 ligand peptidoglycan, a bacterial cell wall component, and the NOD2 ligand muramyl dipeptide, the minimally essential component of peptidoglycan, can also bind to CD14 [20,28] in a manner that is facilitated by LBP [20]. However, pre-mixing HEE with peptidoglycan or muramyl dipeptide did not reduce LPS-induced IL-8 production in dTHP-1 cells (Fig. 3G), suggesting that the ethanol extract does not bind to LBP or CD14. We found that LPS-induced IL-8 production was inhibited in the cells treated with pre-mixed HEE with LPS not only derived from *E. coli* but also derived that from *P. gingivalis* and *F. nucleatum* (Fig. 3F). Since lipid A moieties of *P. gingivalis* and *F. nucleatum* have characteristic structures distinct from those of *E. coli* [29], these results suggest that ergosterol can bind to various lipid A structures. We found that HEE significantly attenuates IL-8 production induced by *E. coli*-derived LPS in dTHP-1 cells at concentrations of 10–100 μg/mL (Fig. 1E). However, we observed no reduction in cell viability when dTHP-1 cells were treated with edible mushroom ethanol extracts at 100 μg/mL (Fig. 1H), indicating that cytotoxicity is not involved in the reduction of IL-8 production. Furthermore, ergosterol peroxide could not suppress the growth of HL-60 cells, a human leukemia myeloid cell line [30]. Overall, these results suggest that the HEE does not attenuate IL-8 production by causing cytotoxicity.

Although sterols are mainly found in ethanol extracts, they are also found in hot water extracts from *H. erinaceus* and other edible mushrooms that have been reported to mainly

contain polysaccharides [22], such as β-glucan [31]. Moreover, water-soluble polysaccharides from *G. frondosa* have been shown to strongly inhibit the production of tumor necrosis factor (TNF)-α and IL-6 induced by LPS in mouse RAW264.7 macrophages [32]. Herein, we found that pre-mixing LPS with the hot water extract of *G. frondosa* mostly attenuates the LPS-induced IL-8, -6, and -1β production in dTHP-1 cells (Fig. 2), suggesting that different compounds may bind to LPS in the ethanol and hot water extracts, leading to attenuation of LPS-induced inflammatory cytokine production. *H. erinaceus*-derived protein (HEP3) can decrease LPS-induced IL-1β, -6, and TNF-α production in mouse RAW264.7 macrophages [8]; thus, future studies should examine the effect of *H. erinaceus* hot water extract on pro-inflammatory cytokine production in host cells.

The gastrointestinal tract is a large reservoir of commensal gram-negative bacteria with LPS in their cell walls. Changes in the microbial composition of the intestinal microbiota, known as dysbiosis, can cause chronic inflammation that leads to the loss of barrier function in intestinal epithelial cells [33]. Therefore, it is important to maintain the balance of intestinal commensal bacteria and prevent inflammation to regulate intestinal homeostasis. Based on our results, we hypothesize that ethanol extracts from edible mushrooms could help prevent inflammatory responses in the intestine. Consistently, previous studies have shown that *H. erinaceus* ethyl acetate extract can attenuate LPS-induced IL-6 and TNF-α production in RAW264.7 cells [13] and that ergosterol peroxide derived from the edible mushroom *Sarcodon aspratus* can inhibit LPS-induced TNF-α production in RAW264.7 cells [25]. Therefore, we believe that edible mushroom extracts could be useful as LPS-neutralizing agents for developing pharmaceuticals and as functional foods for immunotherapy.

Declaration of competing interest

The authors declare no conflicts of interest.

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Figure legends

Fig. 1. Mixture of ethanol extracts from edible mushrooms and lipopolysaccharide (LPS) attenuates pro-inflammatory cytokine production by LPS in dTHP-1 cells. (A) dTHP-1 cells were pretreated with *Hericium erinaceus* ethanol extract (HEE; 100 μg/mL) for 1 h at 37 °C, washed twice with complete medium, and stimulated with *Escherichia coli*-derived LPS (10 ng/mL) for 24 h at 37 °C. (B) dTHP-1 cells were simultaneously stimulated with *E. coli* LPS (10 ng/mL) and HEE (100 μg/mL) for 24 h. (C) *E. coli* LPS (10 ng/mL) was pre-mixed with HEE (100 μg/mL) for 1 h at 37°C, and then dTHP-1 cells were stimulated with the mixture for 24 h. (D-I) *E. coli* LPS (10 ng/mL) was pre-mixed with 100 μg/mL (D, E, F, and H) or the indicated concentrations (G) of ethanol extracts from whole mushroom preparations, or with the indicated concentrations of the lyophilized fruiting body of *H. erinaceus* (I) for 1 h at 37 °C. dTHP-1 cells were stimulated with *E. coli* LPS with or without each mushroom extract for 24 h at 37 °C. (F) To induce mature interleukin (IL)-1β production, cells were stimulated with *E. coli*-derived LPS following treatment with nigericin (5 μM) for 1 h. The concentrations of interleukin (IL)-8 (A–D, G and I), IL-6 (E), and IL-1β (F) in the culture supernatants were analyzed using ELISAs. (H) Viability of dTHP-1 cells treated with 100 μg/mL of each mushroom ethanol extract for 24 h. NanoLuc® substrate luminescence was measured using a luminometer. Viability is expressed as the percentage of viable cells compared to those of the untreated control cells. Data represent the mean ± SD of three independent experiments. Statistically significant differences are indicated as follows: $^{**}p < 0.01$; $^{*}p < 0.05$ *vs.* the respective untreated control.

Fig. 2. Mixture of hot water extracts from edible mushrooms and lipopolysaccharide (LPS) attenuates pro-inflammatory cytokine production by LPS in dTHP-1 cells. *Escherichia coli*-derived LPS (10 ng/mL) was pre-mixed with 100 μg/mL of hot water extract from each whole mushroom preparations for 1 h at 37 °C. dTHP-1 cells were stimulated with *E. coli* LPS in the presence or absence each mushroom extract for 24 h at 37 °C. (C) To induce mature interleukin (IL)-1β production, cells were stimulated with *E. coli* LPS and treated with nigericin (5 μM) for 1 h. The concentrations of IL-8 (A), IL-6 (B), and IL-1β (C) in the culture supernatants were analyzed using ELISAs. Data represent the mean ± SD of three independent experiments. Statistically significant differences are indicated as follows: $^{**}p < 0.01$; $^{*}p < 0.05$ *vs.* the respective untreated control.

Fig. 3. Mixture of ethanol extracts from edible mushrooms and lipid A attenuates interleukin (IL)-8 production by lipid A in dTHP-1 cells. Briefly, 10 ng/mL (A and B) or the indicated concentrations (C) of *Escherichia coli*-type lipid A, or *E. coli*-derived LPS (5 ng/mL) (C) were pre-mixed with 100 μg/mL of ethanol (A) or hot water (B) extracts from each whole mushroom preparation or 100 μg/mL of *Hericium erinaceus* ethanol extract (HEE; C) for 1 h at 37 °C. dTHP-1 cells were stimulated with lipid A or LPS with or without each mushroom extract for 24 h at 37 °C. (D) *E. coli* LPS (10 ng/mL) was pre-mixed with HEE (100 μg/mL) without FBS for 1 h at 37 °C. (E) *E. coli*-derived LPS (20 μg/mL) was pre-mixed with HEE (100 μg/mL) for 1 h at 37 °C. (F and G) HEE (100 μg/mL) was pre-mixed with *Porphyromonas gingivalis*-derived LPS (50 ng/mL) (F), *Fusobacterium nucleatum*-derived LPS (50 ng/mL) (F), *Levilactobacillus brevis*-derived peptidoglycan (10 μg/mL) (G), FSL-1

(100 ng/mL) (G), or muramyl dipeptide (10 μg/mL) (G) for 1 h at 37 °C. dTHP-1 cells (A–D, F and G) or THP-1 cells (E) were stimulated for 24 h (A–G) at 37 °C in the presence (A–C, E–G) or absence (D) of FBS. (A–G) The concentration of IL-8 in the culture supernatants was analyzed using ELISA. Data represent the mean ± SD of three independent experiments. Statistically significant differences are indicated as follows: $^{**}p < 0.01$; $^{*}p < 0.05$ *vs.* the respective untreated control.

Fig. 4. Ethanol extract from *Hericium erinaceus* containing ergosterol, and binding of ergosterol and related compounds to LPS decrease IL-8 production in dTHP-1 cells. (A) *Hericium erinaceus* ethanol extract (HEE) (100 μg/mL) was preheated for 30 min at 95 °C and pre-mixed with *Escherichia coli*-derived LPS (10 ng/mL) for 1 h at 37 °C. (D and E) *E. coli* LPS (10 ng/mL) (D and E), *Porphyromonas gingivalis*-derived LPS (50 ng/mL) (D), or *Fusobacteriuim nucleatum*-derived LPS (50 ng/mL) (D) was pre-mixed with ergosterol standard (100 μg/mL) (D), ergosterol peroxide (100 μg/mL) (E), or ercalcitriol (100 μg/mL) (E) for 1 h at 37 °C. (A, D, and E) dTHP-1 cells were stimulated with LPS with or without HEE, ergosterol, ergosterol peroxide, or ercalcitriol for 24 h at 37 °C. The concentrations of IL-8 in the culture supernatants were analyzed using ELISA. Data represent the mean ± SD of three independent experiments. Statistically significant differences are indicated as follows: $^{**}p < 0.01$ *vs.* the respective untreated control. (B) TLC determination of ergosterol and HEE components. The plate was developed with a solvent system of chloroform/methanol/acetate/water (90:10:1:1, v/v/v/v). Spots were visualized by spraying

10% H_2SO_4/ethanol and heating. (C) Gas chromatography-mass spectrometry chromatogram of ergosterol and HEE components.

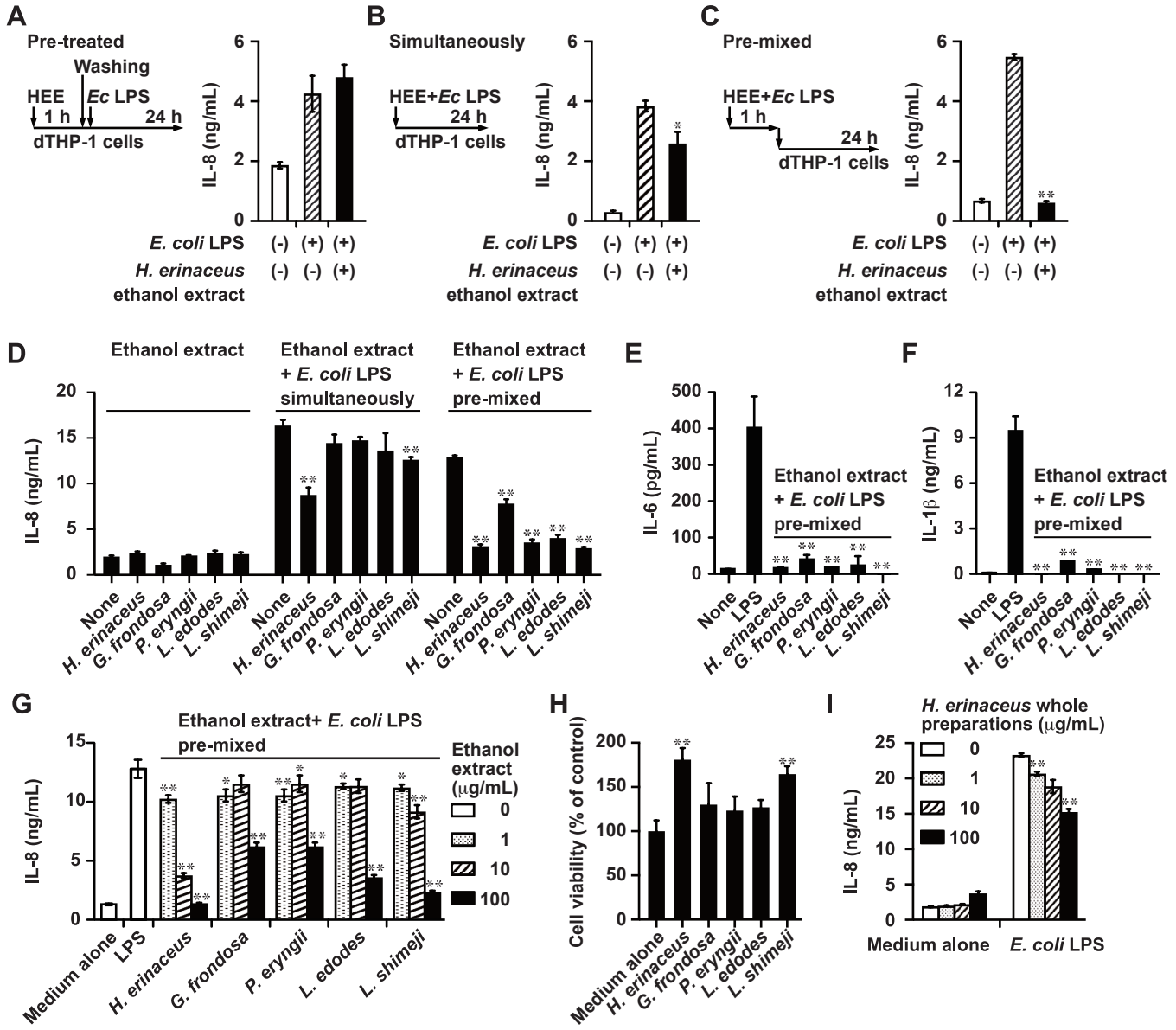
A
Pre-treated
Washing
HEE
Ec LPS
1 h
24 h
dTHP-1 cells
IL-8 (ng/mL)
E. coli LPS (-) (+) (+)
H. erinaceus ethanol extract (-) (-) (+)
B
Simultaneously
HEE+Ec LPS
24 h
dTHP-1 cells
IL-8 (ng/mL)
E. coli LPS (-) (+) (+)
H. erinaceus ethanol extract (-) (-) (+)
C
Pre-mixed
HEE+Ec LPS
1 h
24 h
dTHP-1 cells
IL-8 (ng/mL)
E. coli LPS (-) (+) (+)
H. erinaceus ethanol extract (-) (-) (+)
D
Ethanol extract
Ethanol extract + E. coli LPS simultaneously
Ethanol extract + E. coli LPS pre-mixed
IL-8 (ng/mL)
None
H. erinaceus
G. frondosa
P. eryngii
L. edodes
L. shimeji
E
IL-6 (pg/mL)
Ethanol extract + E. coli LPS pre-mixed
None
LPS
H. erinaceus
G. frondosa
P. eryngii
L. edodes
L. shimeji
F
IL-1β (ng/mL)
Ethanol extract + E. coli LPS pre-mixed
None
LPS
H. erinaceus
G. frondosa
P. eryngii
L. edodes
L. shimeji
G
Ethanol extract+ E. coli LPS pre-mixed
IL-8 (ng/mL)
Ethanol extract (μg/mL)
0
1
10
100
Medium alone
LPS
H. erinaceus
G. frondosa
P. eryngii
L. edodes
L. shimeji
H
Cell viability (% of control)
Medium alone
H. erinaceus
G. frondosa
P. eryngii
L. edodes
L. shimeji
I
H. erinaceus whole preparations (μg/mL)
0
1
10
100
IL-8 (ng/mL)
Medium alone
E. coli LPS

Figure 1

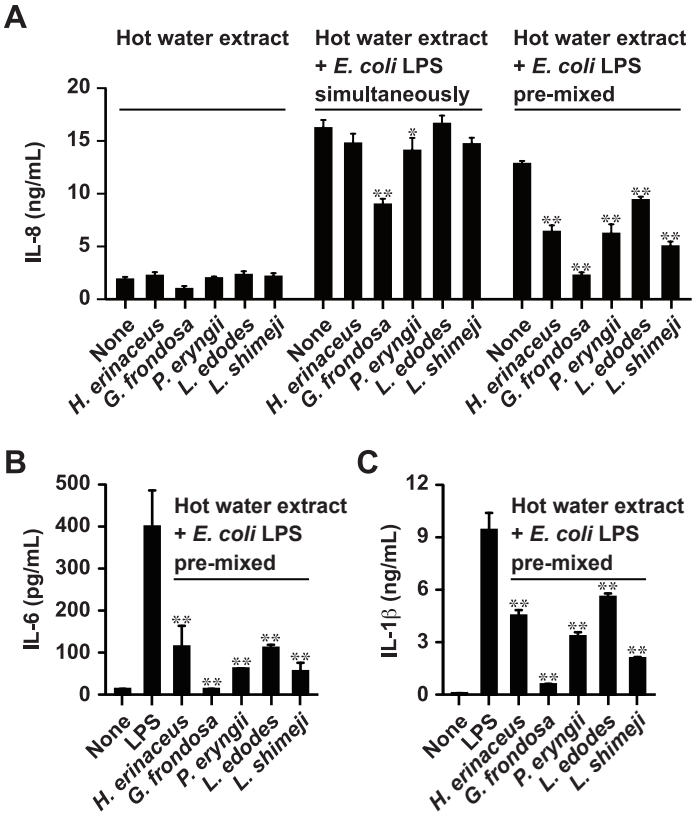
A
Hot water extract
Hot water extract
+ E. coli LPS
simultaneously
Hot water extract
+ E. coli LPS
pre-mixed
IL-8 (ng/mL)
20
15
10
5
0
None
H. erinaceus
G. frondosa
P. eryngii
L. edodes
L. shimeji
B
IL-6 (pg/mL)
500
400
300
200
100
0
Hot water extract
+ E. coli LPS
pre-mixed
None
LPS
H. erinaceus
G. frondosa
P. eryngii
L. edodes
L. shimeji
C
IL-1β (ng/mL)
12
9
6
3
0
Hot water extract
+ E. coli LPS
pre-mixed
None
LPS
H. erinaceus
G. frondosa
P. eryngii
L. edodes
L. shimeji

Figure 2

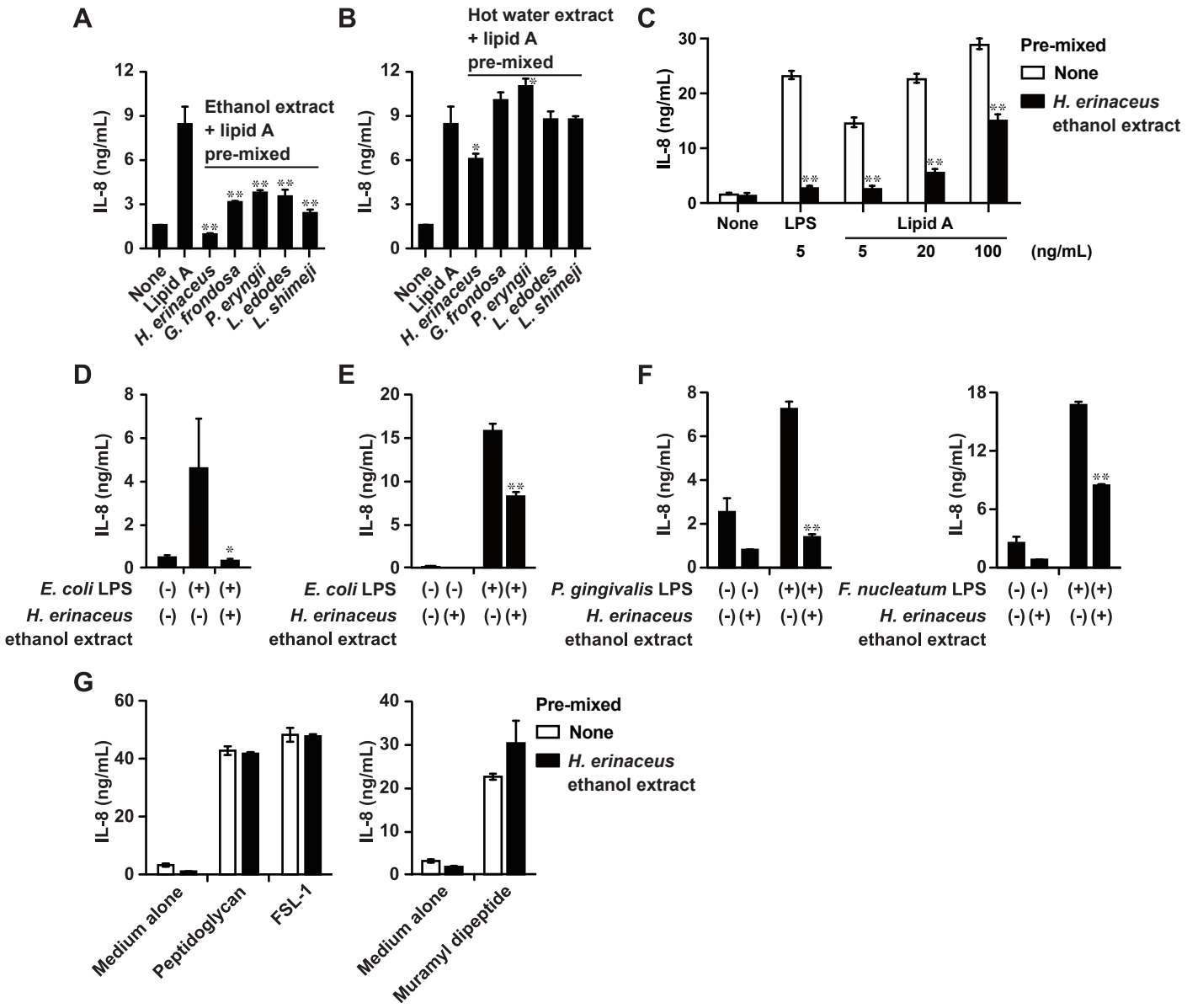
A
IL-8 (ng/mL)
Ethanol extract
+ lipid A
pre-mixed
None
Lipid A
H. erinaceus
G. frondosa
P. eryngii
L. edodes
L. shimeji
B
Hot water extract
+ lipid A
pre-mixed
C
Pre-mixed
None
H. erinaceus
ethanol extract
Lipid A
(ng/mL)
D
E. coli LPS
H. erinaceus
ethanol extract
E
P. gingivalis LPS
F
F. nucleatum LPS
G
Medium alone
Peptidoglycan
FSL-1
Muramyl dipeptide

Figure 3

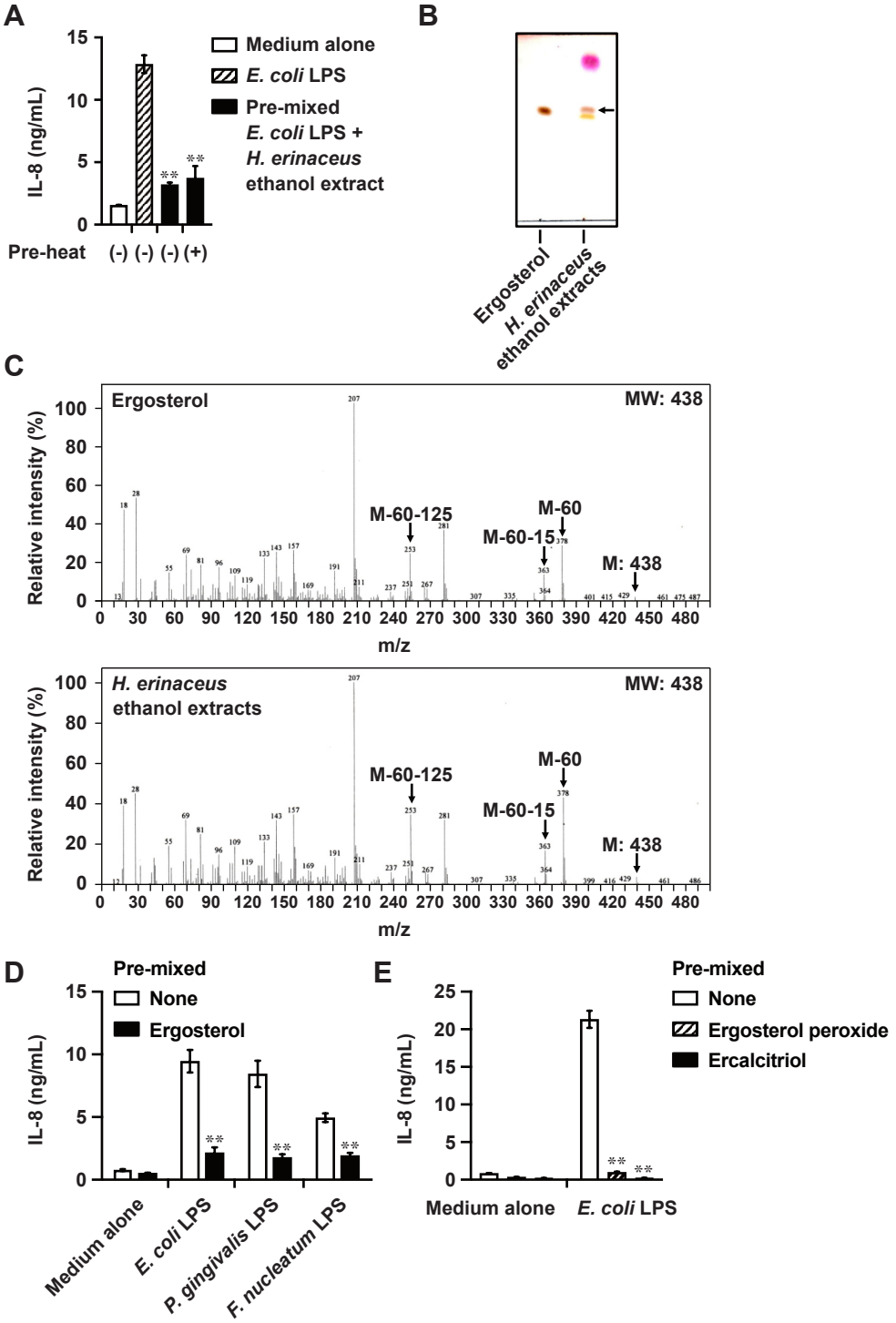
A
IL-8 (ng/mL)
Medium alone
E. coli LPS
Pre-mixed
E. coli LPS +
H. erinaceus
ethanol extract
Pre-heat (-) (-) (-) (+)
B
Ergosterol
H. erinaceus
ethanol extracts
C
Ergosterol
MW: 438
Relative intensity (%)
M-60-125
M-60
M-60-15
M: 438
m/z
H. erinaceus
ethanol extracts
MW: 438
Relative intensity (%)
M-60-125
M-60
M-60-15
M: 438
m/z
D
Pre-mixed
None
Ergosterol
IL-8 (ng/mL)
Medium alone
E. coli LPS
P. gingivalis LPS
F. nucleatum LPS
E
Pre-mixed
None
Ergosterol peroxide
Ercalcitriol
IL-8 (ng/mL)
Medium alone
E. coli LPS

Figure 4